

Homework 7

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Homework

7

ECE 3340 Numerical Methods

Homework 7: Precision Loss

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Problem 1: Bisection Method

Calculate $\cos^2(x) = e^x$ using 5 iterations of the bisection method on the interval $[-3, -1]$.

	a	f(a)	b	f(b)	c	f(c)
1	<i>-3.0000</i>	<i>0.930298</i>	<i>-1.6660</i>	<i>-0.075953</i>	<i>-2.0000</i>	<i>0.037843</i>
2	<i>-2.0000</i>	<i>0.037843</i>	<i>-1.0000</i>	<i>-0.075953</i>	<i>-1.5000</i>	<i>-0.218126</i>
3	<i>-2.0000</i>	<i>0.037843</i>	<i>-1.5000</i>	<i>-0.218126</i>	<i>-1.7500</i>	<i>-0.142002</i>
4	<i>-2.0000</i>	<i>0.037843</i>	<i>-1.7500</i>	<i>-0.142002</i>	<i>-1.8750</i>	<i>-0.063635</i>
5	<i>-2.0000</i>	<i>0.037843</i>	<i>-1.8750</i>	<i>-0.063635</i>	<i>-1.9375</i>	<i>-0.015513</i>

$$f'(x) = \frac{d}{dx}(\cos^2(x) - e^x) = -\sin(2x) - e^x$$

$$x = -1.9375$$

Problem 2: Newton's Method

Calculate $\cos^2(x) = e^x$ using 5 iterations of Newton's method with $x_0 = -3$.

	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
1	<i>-3.0000</i>	<i>0.930298</i>	<i>-0.329203</i>	<i>-0.174086...</i>
2	<i>-0.174086...</i>	<i>0.129774</i>	<i>-0.499044</i>	<i>0.0859598...</i>
3	<i>0.085959886</i>	<i>-0.097134</i>	<i>-1.260837</i>	<i>0.008920941</i>

3	0.085959886	-0.097134	-1.260837	0.008920941
4	0.008920941	-0.009040	-1.026802	0.000116484
5	0.000116484	-0.000116504	-1.00349	2.03×10^{-8}

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x = 7.03 \times 10^{-8}$$

